Design and Deployment of 3 -Tier Web Application on AWS Cloud

Introduction

This project demonstrates the implementation of a 3-tier architecture using Amazon Web Services (AWS). The architecture is divided into presentation, application, and database layers to ensure better security, scalability, and performance. Various AWS services such as VPC, EC2, Load Balancer, and RDS are used to build a highly available system. This project provides hands-on experience in designing real-world cloud infrastructure.

Objective

To design and deploy a scalable and secure 3-tier architecture using AWS cloud services.

To gain hands-on experience in implementing real-world cloud infrastructure and best practices .

Purpose

To gain practical knowledge of AWS services and implement real-world cloud architecture.

Architecture Overview

The project is designed using a 3-tier architecture model, which separates the application into three layers: Presentation Layer, Application Layer, and Database Layer.

This structure improves scalability, security, and maintainability

AWS Services Used

- **EC2 (Elastic Compute Cloud)** – Used to host frontend and backend servers
- **VPC (Virtual Private Cloud)** – Created network with public & private subnets
- **RDS (Relational Database Service)** – Used for database (MySQL/MariaDB)
- **S3 (Simple Storage Service)** – Used for storing static files (optional)
- **IAM (Identity and Access Management)** – Managed user access and permissions
- **Application Load Balancer** – Distributed traffic across servers
- **Auto Scaling** – Automatically scales instances based on load
- **Security Groups** – Controlled inbound and outbound traffic

AWS Services Used in 3-Tier Architecture

Networking

- VPC
- Public Subnet
- Private Subnet
- Internet Gateway
- NAT Gateway
- Route Tables

Presentation Layer (Frontend)

- Application Load Balancer (ALB)

Application Layer (Backend)

- EC2 Instances
- Auto Scaling Group
- Security Groups

Database Layer

- RDS (MySQL / MariaDB)

Security & Access

- IAM (Users, Roles, Policies)
- Security Groups

Implementation Steps

Create VPC

A Virtual Private Cloud (VPC) was created to establish a secure and isolated network environment in AWS. A custom CIDR block (10.0.0.0/16) was assigned to define the IP address range for the resources.

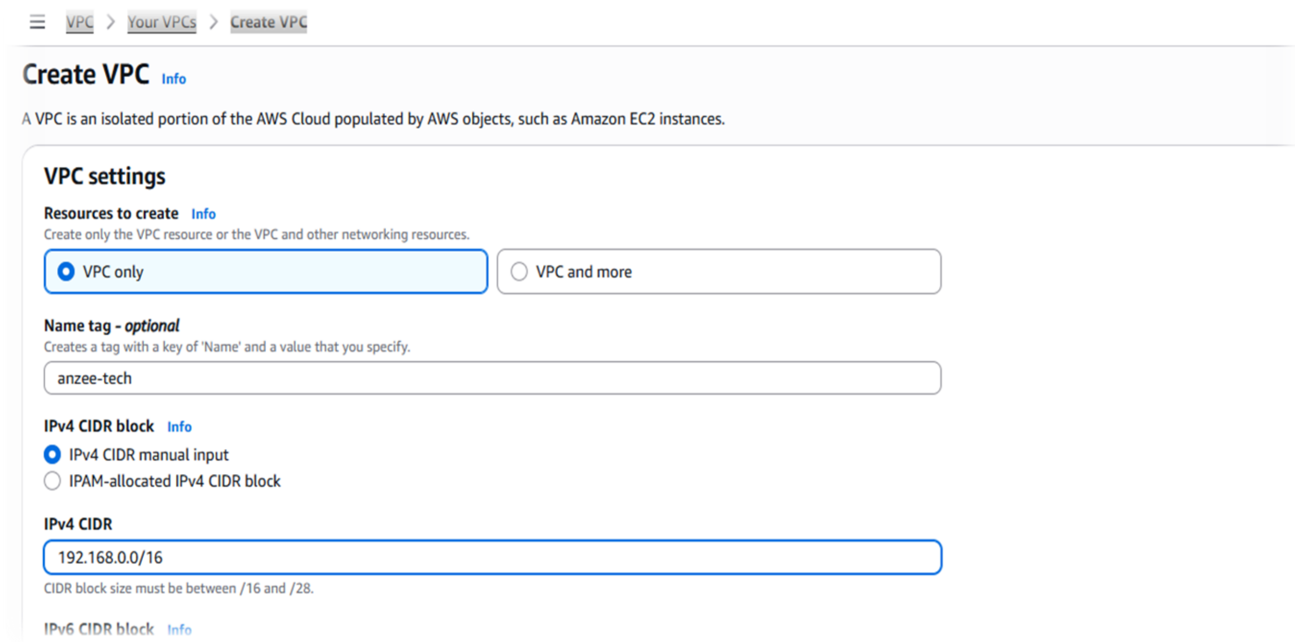


Figure: VPC Creation in AWS

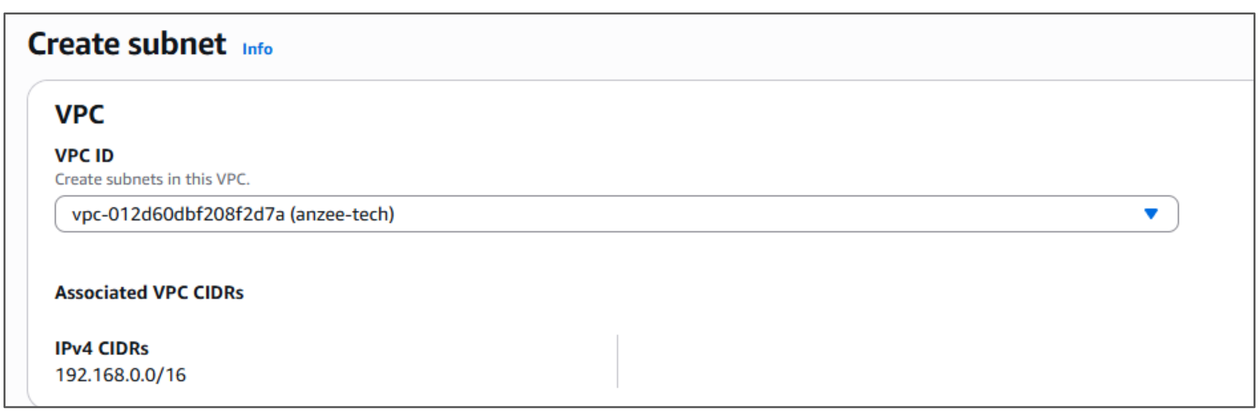
VPC name

CIDR Block

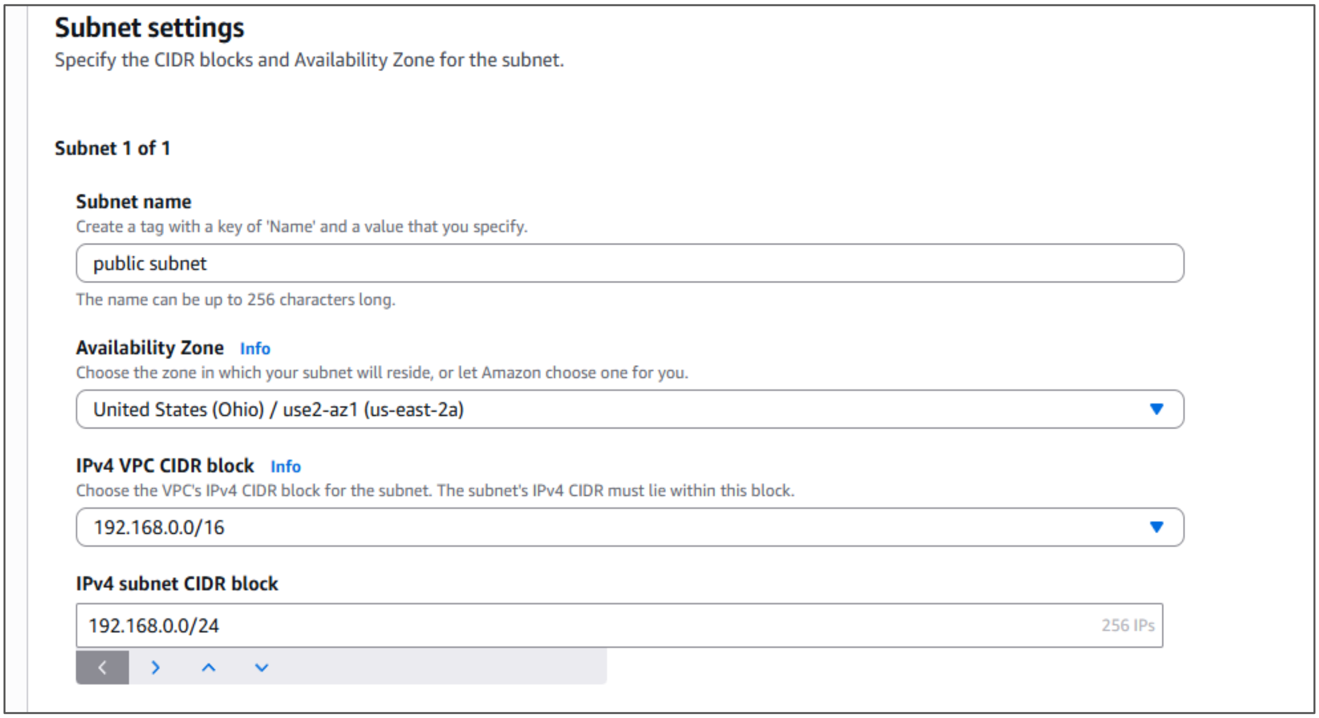
Create Subnets

Subnets were created within the VPC to divide the network into smaller sections. Public subnets were used for internet-facing resources like the Load Balancer, while private subnets were used for application servers and the database to enhance security and control access.

VPC ID

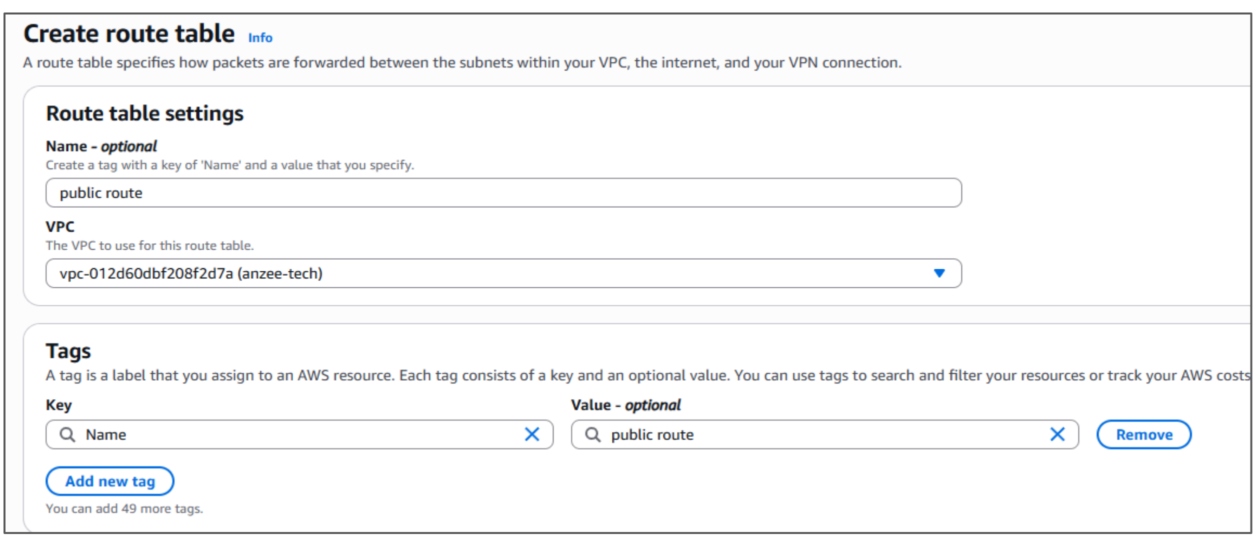


Subnet Name, CIDR block, Availability Zone



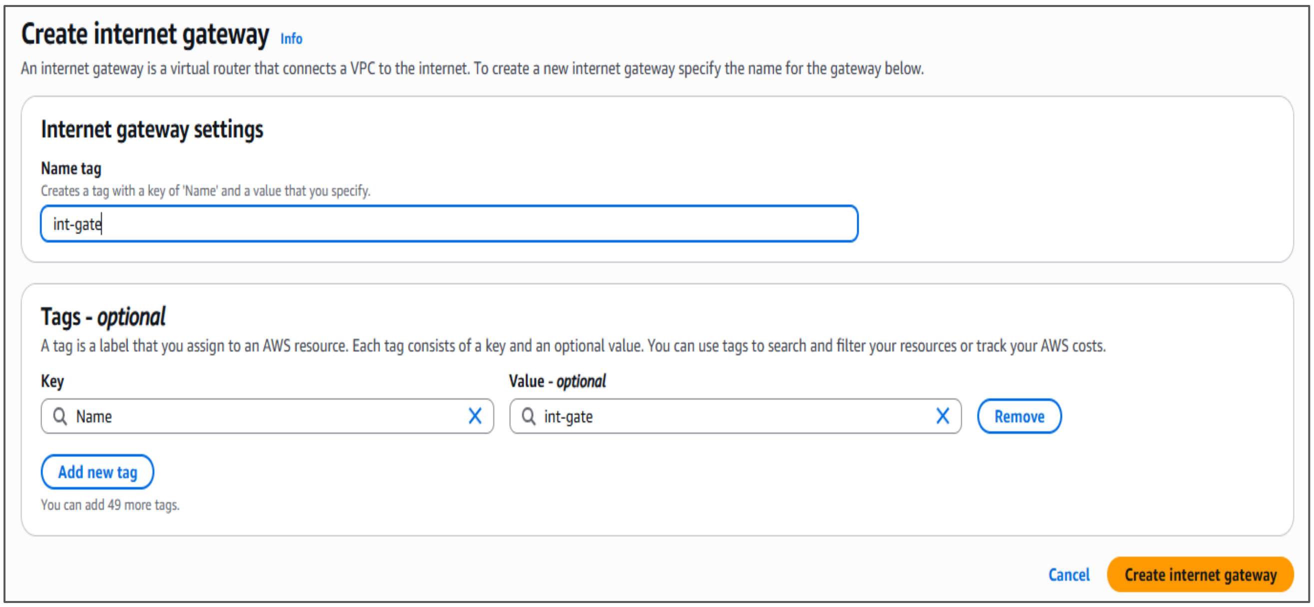
Create Route Tables

Route tables were created to control the flow of network traffic within the VPC. A public route table was configured with a route to the Internet Gateway to allow external access. A private route table was configured to route traffic through a NAT Gateway, enabling secure outbound internet communication. Each subnet was associated with the appropriate route table.



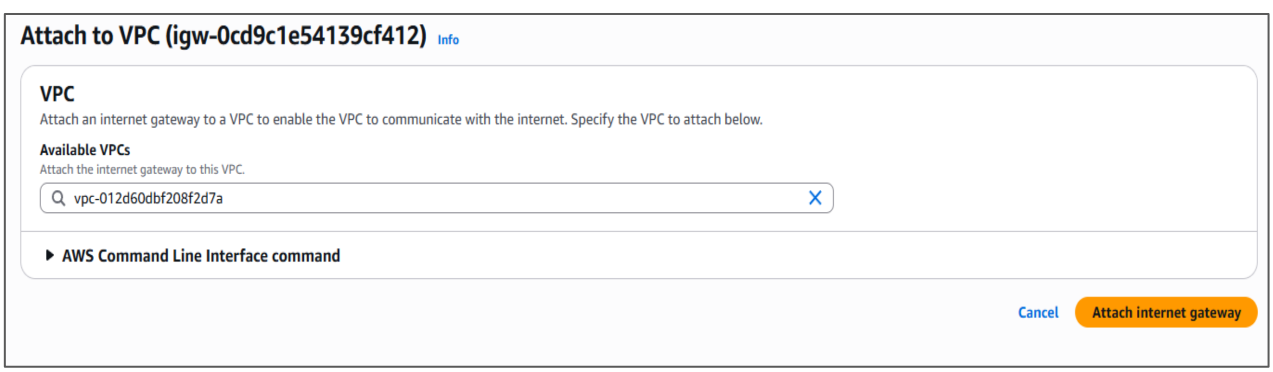
Create Internet Gateway

An Internet Gateway was created and attached to the VPC to enable communication between resources in the VPC and the internet. A route was added in the public route table to allow internet access for resources in the public subnet.



Internet gateway name

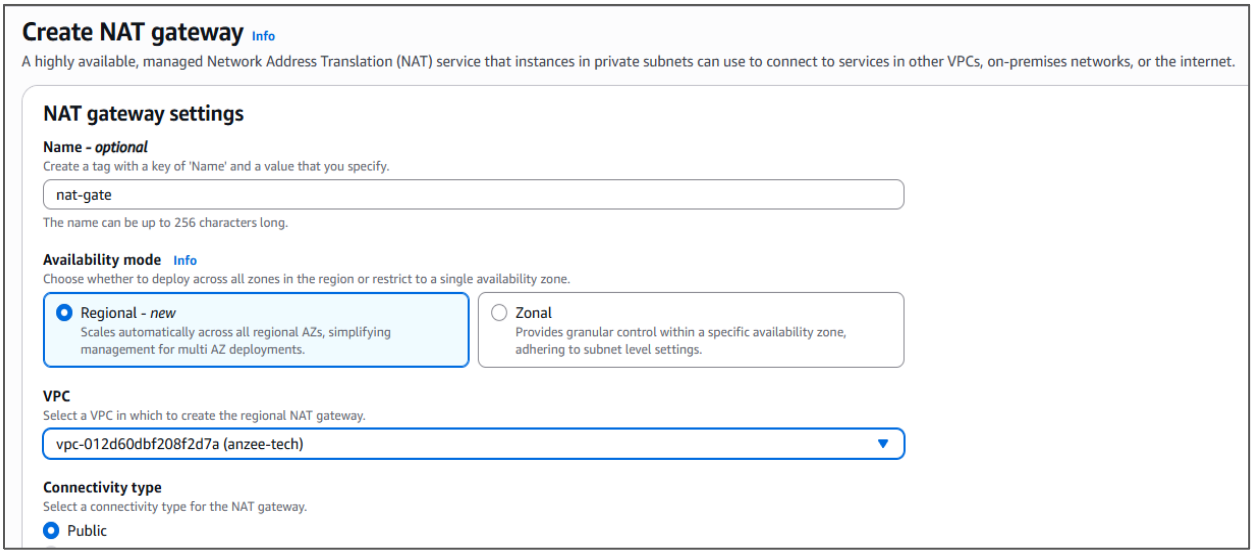
Attach to VPC



Create NAT Gateway

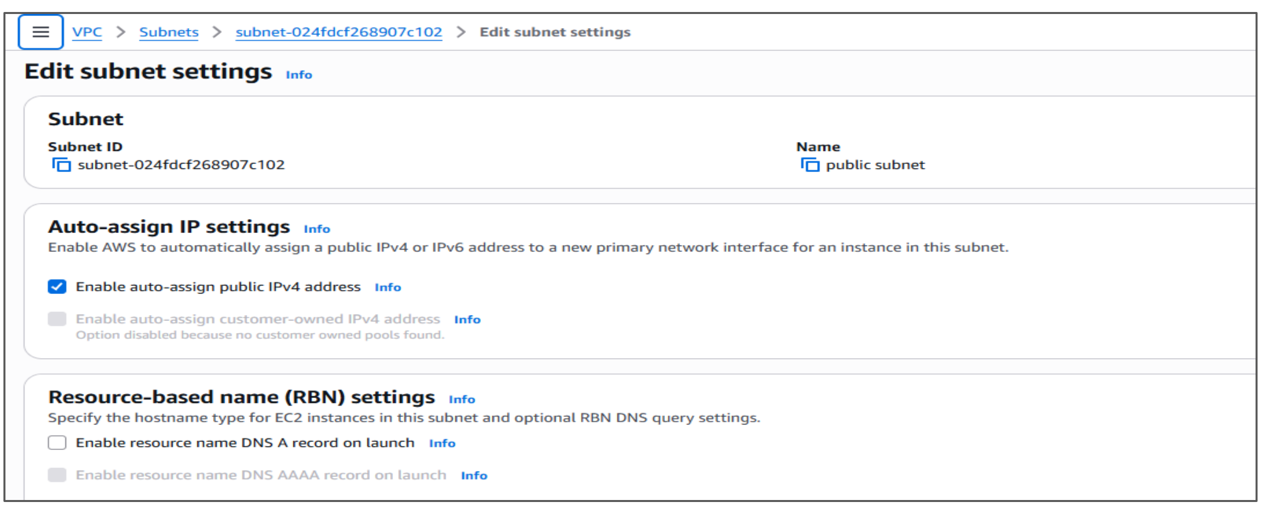
A NAT Gateway was created in the public subnet to allow instances in the private subnet to access the internet securely. It enables outbound internet communication while preventing direct inbound access from external sources.

NAT Gateway created in public subnet for private internet access.

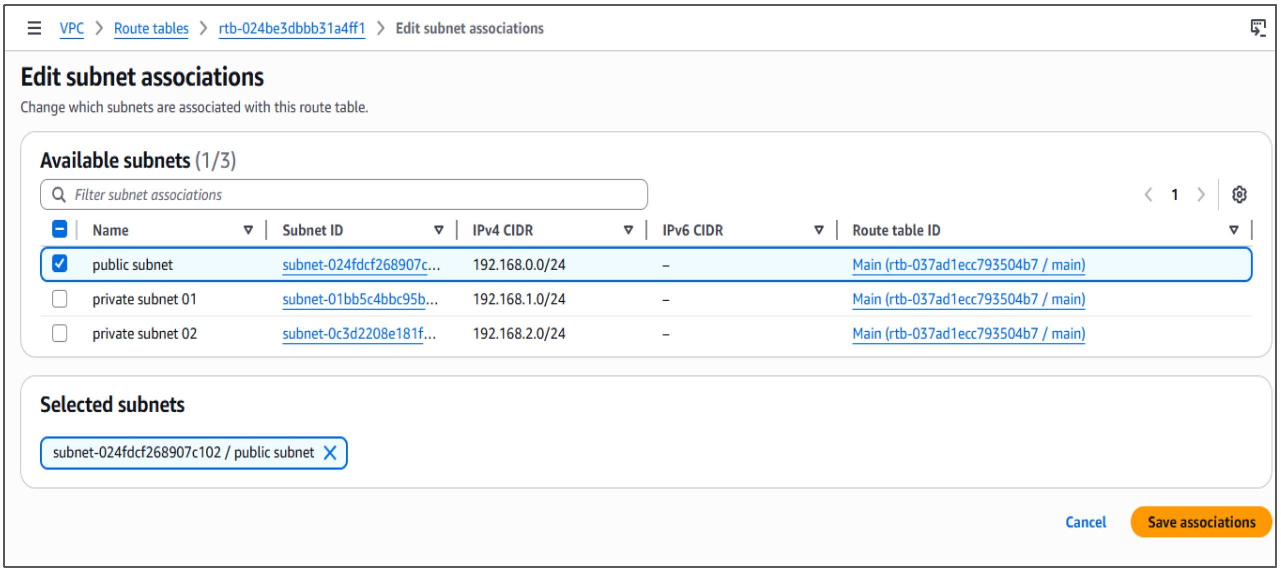


Public Subnet – Auto Assign Public IP

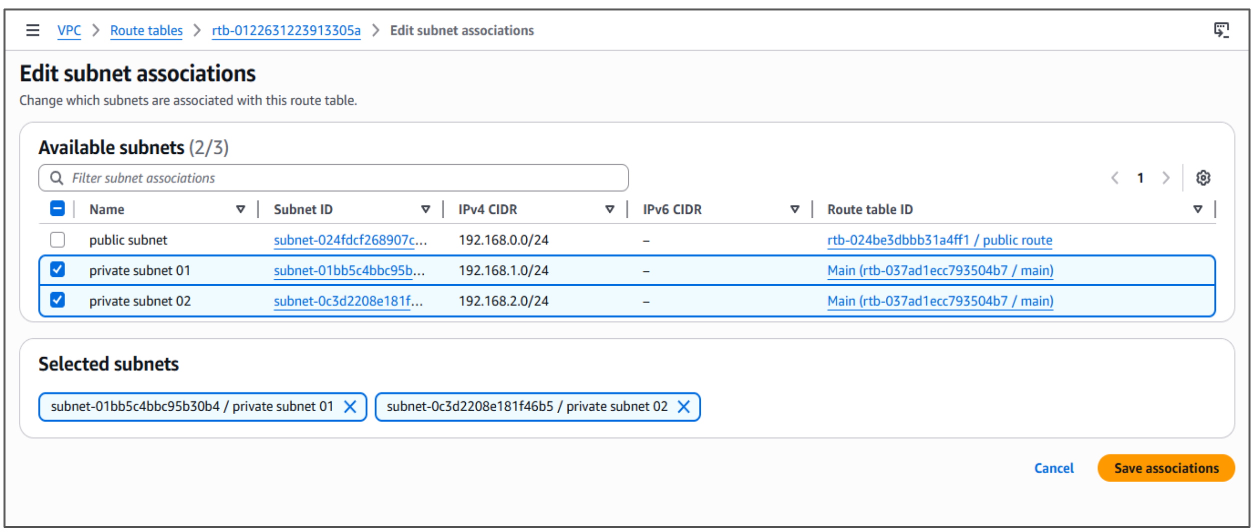
Auto-assign Public IP was enabled for the public subnet to automatically assign a public IP address to instances launched within it, allowing direct internet access.



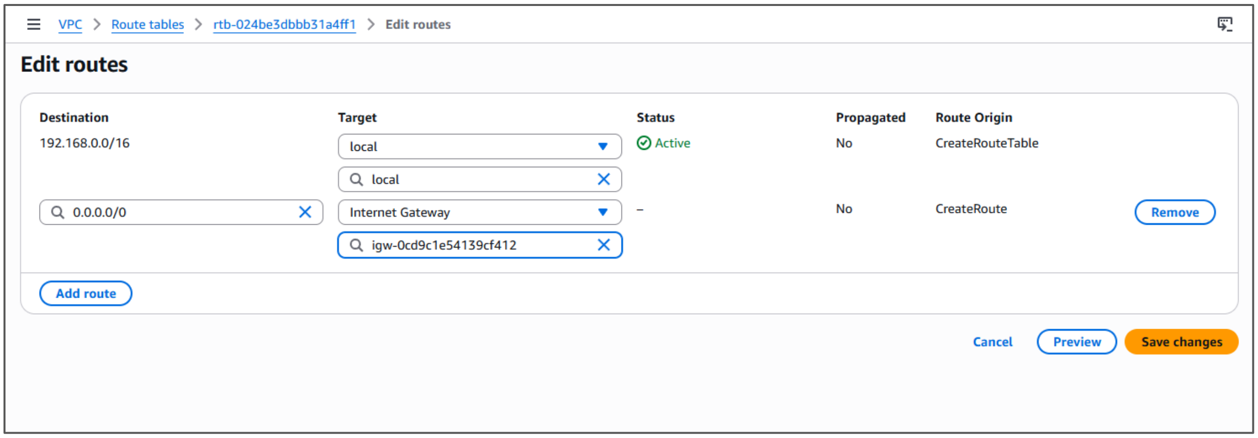
Public Route Table – Subnet Association



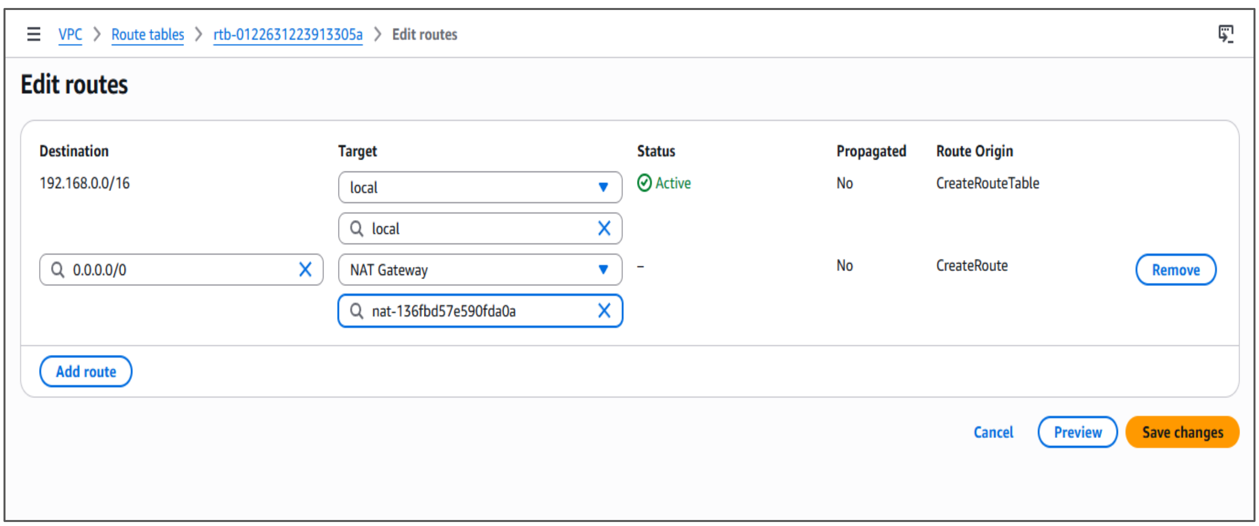
Private Route Table – Subnet Association



Add internet gateway in public route



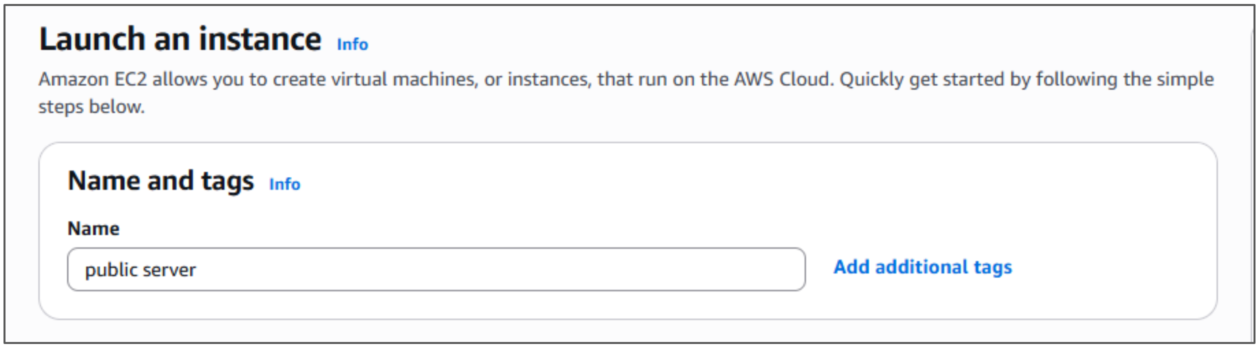
Add NAT gateway in private route



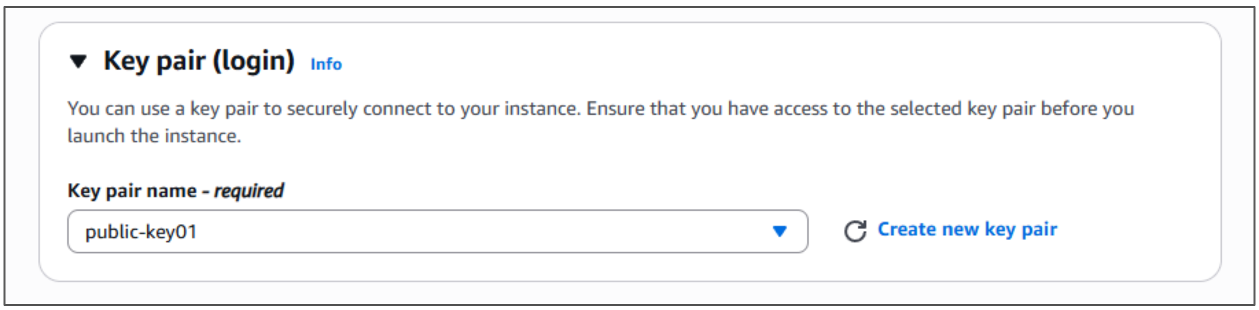
Launch EC2 Instances

EC2 instances were launched in private subnets to host the application. These instances handle the business logic and process user requests coming from the load balancer.

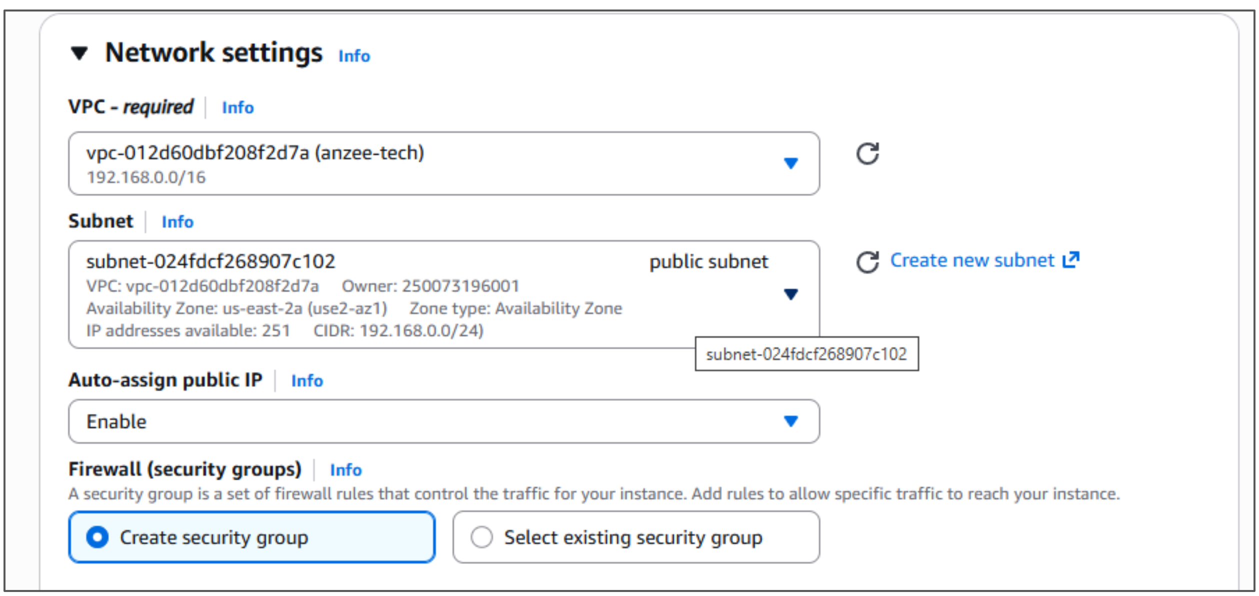
Public server



Name

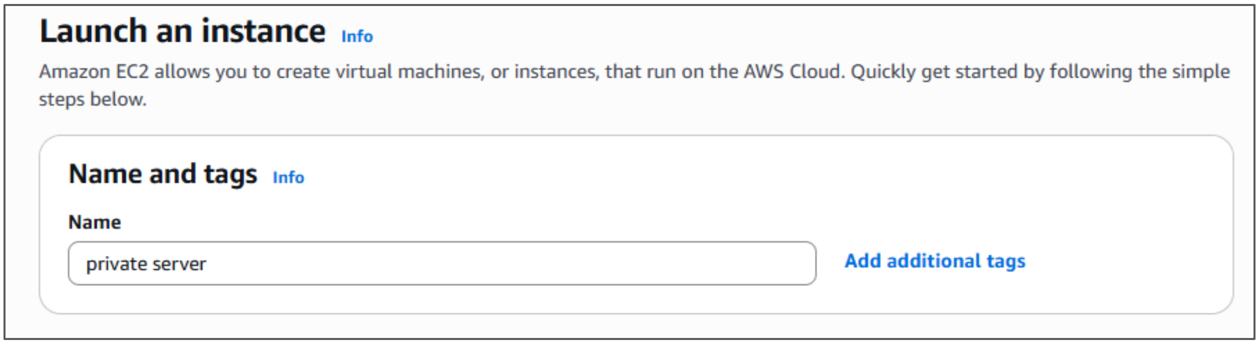


Create key pair

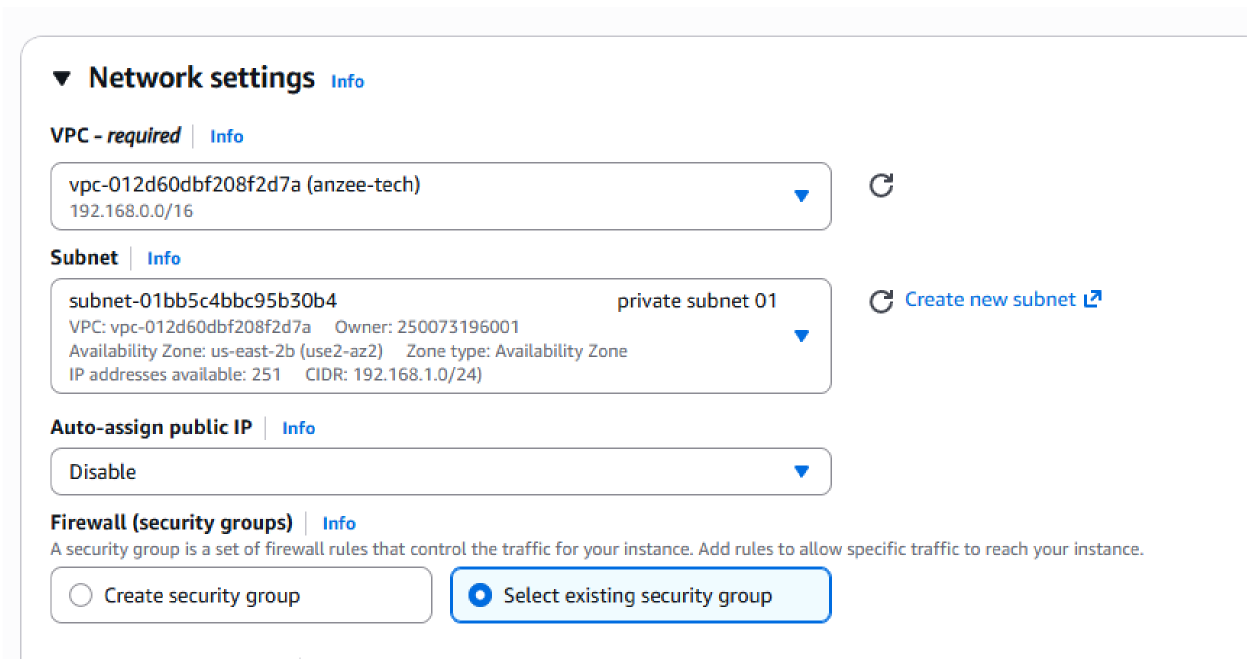


VPC ID, Public Subnet, Create security group

Private server



Name



VPC ID, Private subnet 01, existing security group

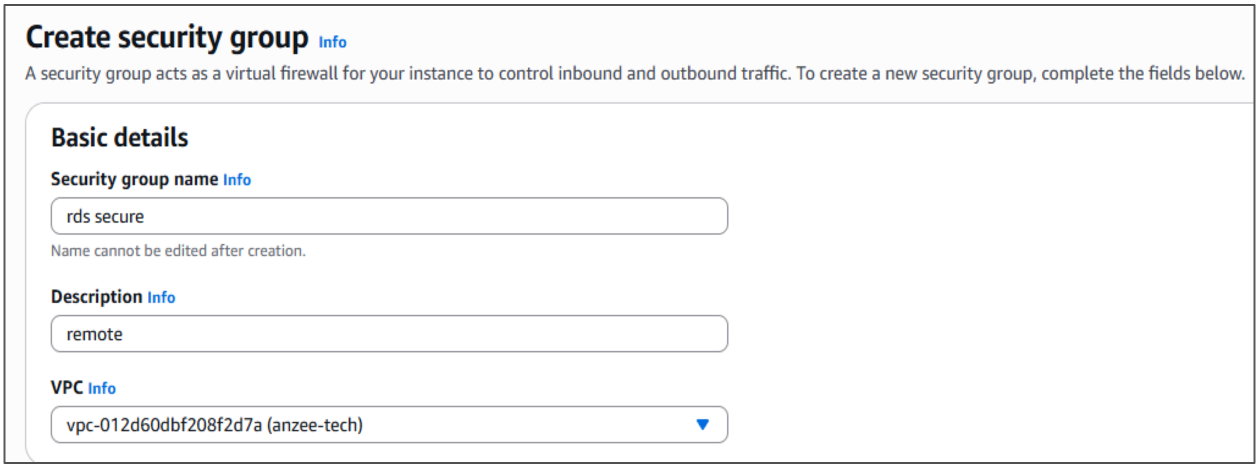
Public connection

```
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Apr  9 11:42:34 2026 from 3.16.146.5
[ec2-user@ip-192-168-0-45 ~]$ sudo passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-192-168-0-45 ~]$ su
Password:
[root@ip-192-168-0-45 ec2-user]# pwd
/home/ec2-user
[root@ip-192-168-0-45 ec2-user]# nano public-key01.pem
[root@ip-192-168-0-45 ec2-user]# ls
public-key01.pem
[root@ip-192-168-0-45 ec2-user]# chmod 400 "public-key01.pem"
[root@ip-192-168-0-45 ec2-user]# ssh -i "public-key01.pem" ec2-user@192.168.1.235
The authenticity of host '192.168.1.235 (192.168.1.235)' can't be established.
ED25519 key fingerprint is SHA256:3AfxdOvg+FWPwVuRzSYDEzxg9/kBod9wwntwTtCoKrk.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.235' (ED25519) to the list of known hosts.
```

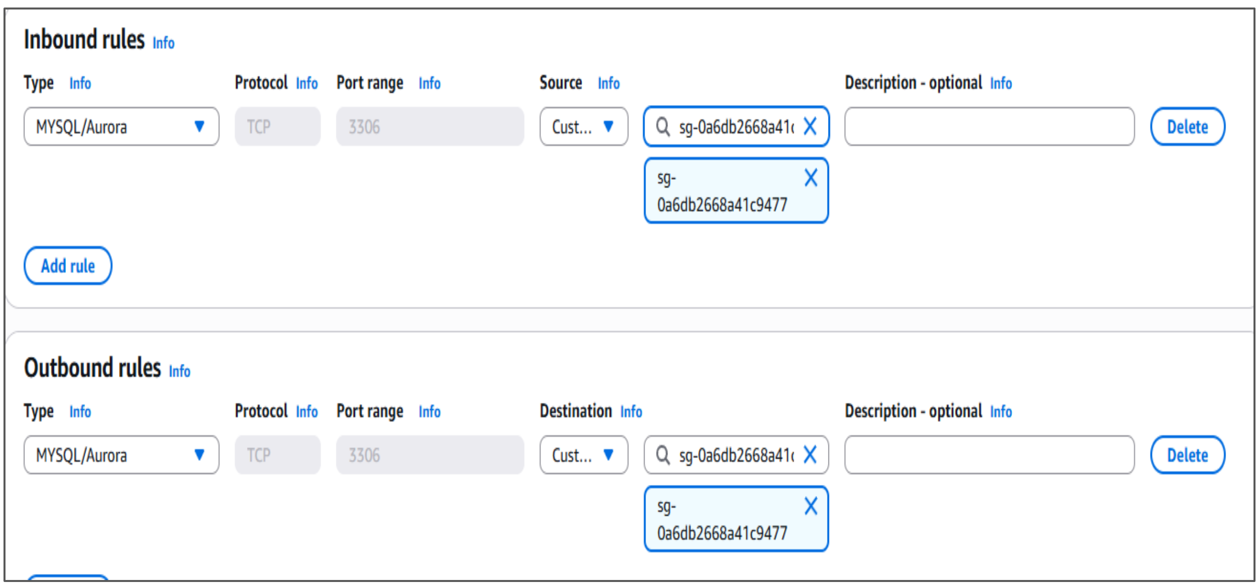
Private connection

```
[root@ip-192-168-0-45 ec2-user]# ssh -i "public-key01.pem" ec2-user@192.168.1.235
The authenticity of host '192.168.1.235 (192.168.1.235)' can't be established.
ED25519 key fingerprint is SHA256:3AfxdOvg+FWPwVuRzSYDEzxg9/kBod9wwntwTtCoKrk.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.235' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-192-168-1-235 ~]$ sudo passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-192-168-1-235 ~]$ su
Password:
[root@ip-192-168-1-235 ec2-user]#
```

Create new security group



Name, description, VPC ID

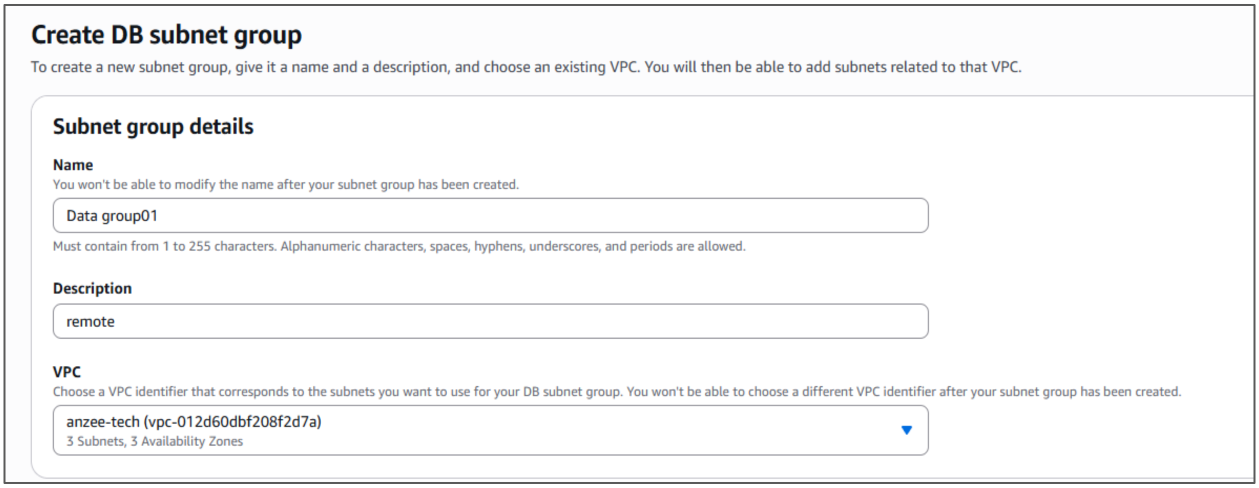


Inbound- MySQL/Aurora Source- security group (private server)

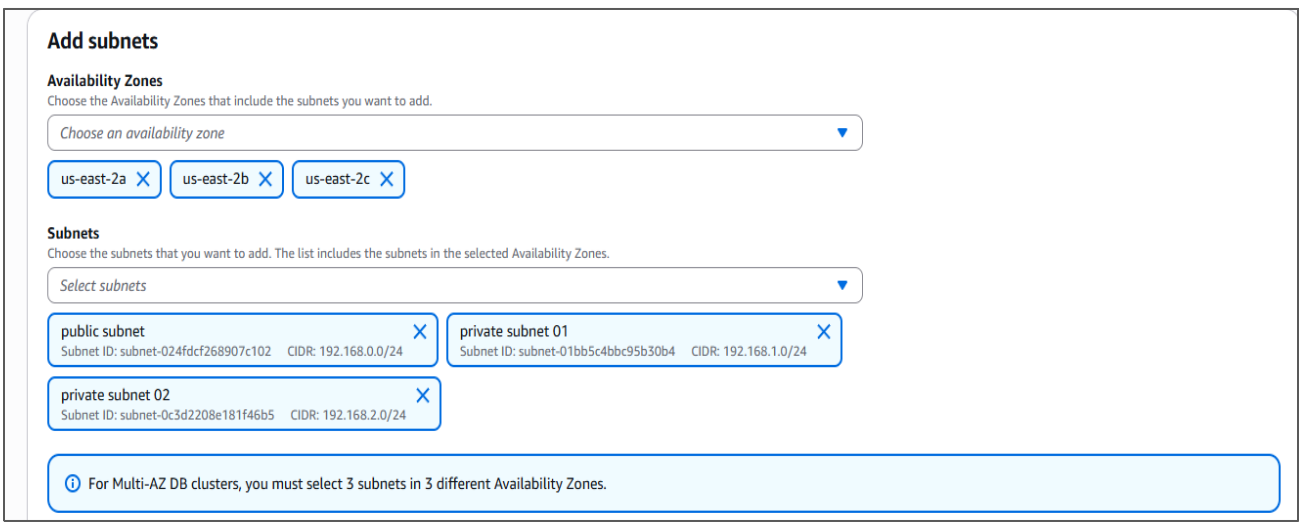
Outbound- MySQL/Aurora Source- security group (private server)

RDS

Create Subnet group



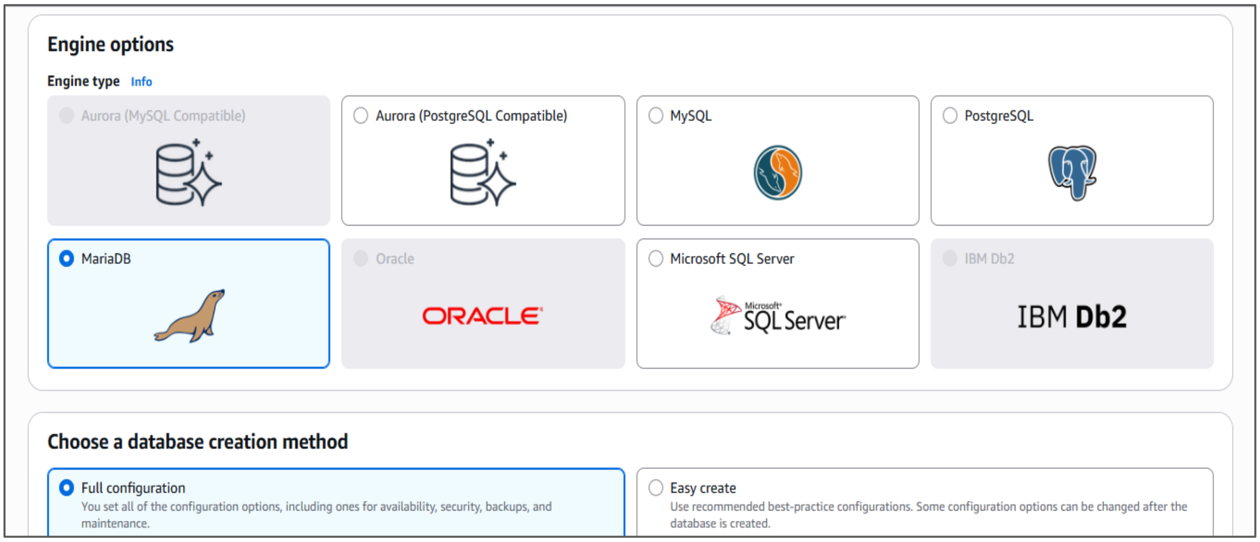
Name, description, VPC ID



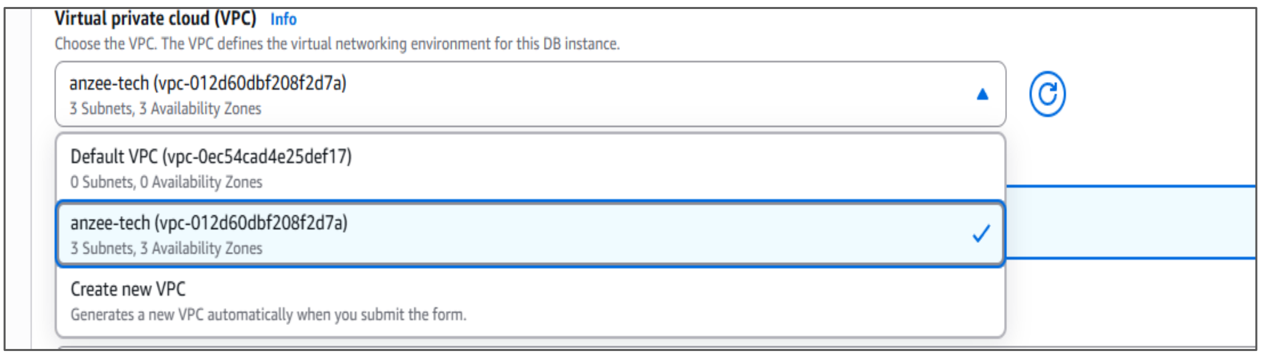
Availability Zones (Three), Subnets (Three)

Create

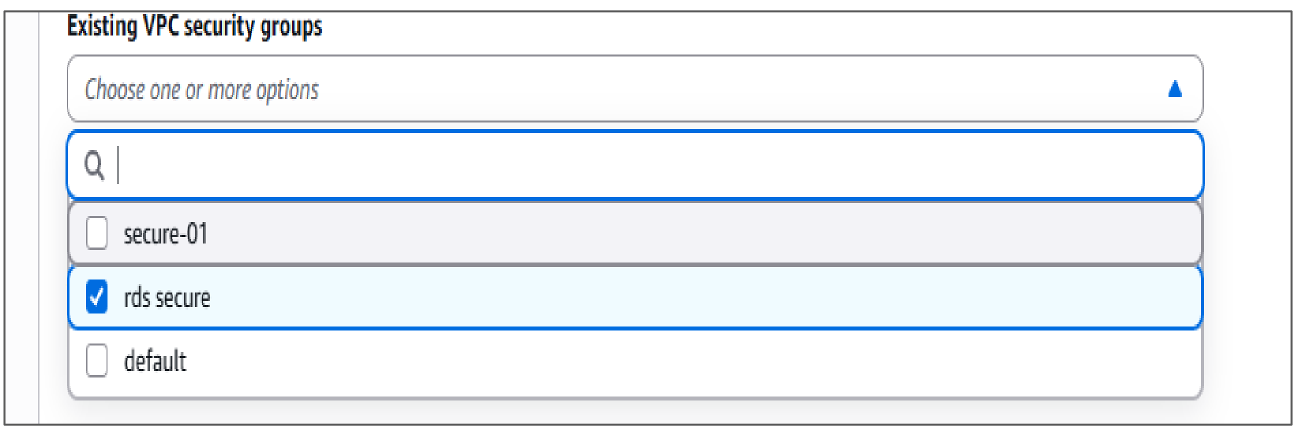
Create databases



MariaDB, Full configuration



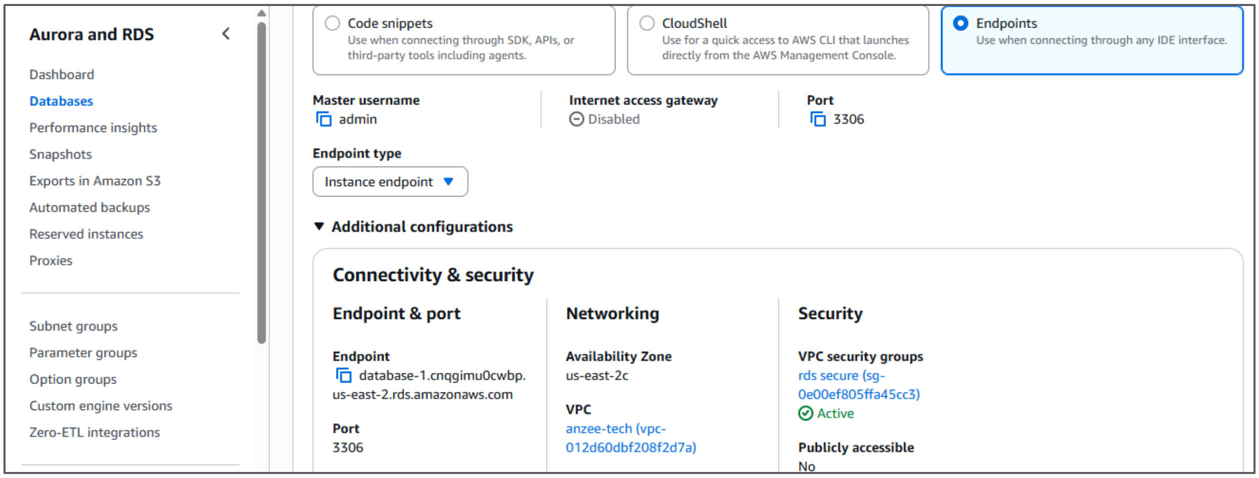
VPC ID



New security group

Create

Endpoint RDS



Connection

```
[ec2-user@ip-192-168-0-45 ~]$ sudo passwd root
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[ec2-user@ip-192-168-0-45 ~]$ su
Password:
[root@ip-192-168-0-45 ec2-user]# dnf install mariadb105-server -y
Last metadata expiration check: 0:05:01 ago on Thu Apr  9 13:06:20 2026.
Package mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-192-168-0-45 ec2-user]# mariadb -hdatabase-1.cnqgimu0cwbp.us-east-2.rds.amazonaws.com -P 3306 -u admin -p --ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 87
Server version: 11.8.6-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Table created

```
MariaDB [school]> SHOW TABLES;
+-------------------+
| Tables_in_school |
+-------------------+
| students          |
+-------------------+
1 row in set (0.004 sec)

MariaDB [school]> INSERT INTO students (name, age) VALUES ('vikram', 21),('surya', 22),('vasu', 25);
Query OK, 3 rows affected (0.012 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [school]> SELECT * FROM students;
+----+--------+------+
| id | name   | age  |
+----+--------+------+
|  1 | vikram |   21 |
|  2 | surya  |   22 |
|  3 | vasu   |   25 |
+----+--------+------+
3 rows in set (0.002 sec)

MariaDB [school]>
```